

Masoud Hadi

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RESEARCH INTERESTS

- ◊ Reinforcement Learning Theory
- ◊ Multi-Agent Reinforcement Learning
- ◊ Learning in Games
- ◊ Machine Learning Theory

EDUCATION

- Isfahan University** Isfahan, Iran
 - *M.Sc. in Computer Engineering – Artificial Intelligence* 2024-Present
 - GPA: 18.22/20, Ranked 2/17*
 - Research Focus: Multi-Agent Reinforcement Learning, Markov Games, and Learning in Games*
- Isfahan University of Technology** Isfahan, Iran
 - *B.Sc. in Computer Engineering – Intelligent Systems* 2019-2024
 - GPA: 17.96/20, Last Two-Year GPA: 18.22/20, Rank: 9/80*
 - Thesis: Community Detection with Graph Neural Networks; Grade: 20/20*

RESEARCH EXPERIENCE

- Isfahan University** Isfahan, Iran
 - *M.Sc. Research – Multi-Agent Reinforcement Learning* 2026 – Present
 - Developing scalable equilibrium-computation methods for finite-horizon factored Markov games by exploiting local transition/reward structure and networked strategic interactions.
 - Deriving structural conditions under which Bellman Q -functions admit local or pairwise decompositions, with the goal of avoiding enumeration of the full joint state-action space.
 - Investigating Nash and coarse correlated equilibrium computation together with computational and statistical complexity guarantees in structured multi-agent settings.
- EPFL (École polytechnique fédérale de Lausanne)** Remote
 - *Under the supervision of Prof. Mackenzie W. Mathis.* Feb 2025 – Jul 2025
 - Contributed to **DISTIL** (ICCV 2025), a data-free trigger-inversion framework that uses latent diffusion to reconstruct suspicious Trojan/backdoor triggers.
 - Investigated internal attention representations and intrinsic dimensionality as signals for detecting backdoored neural networks.
- Research Collaboration with S-Lab, Nanyang Technological University** Remote
 - *Under the supervision of Prof. Ziwei Liu and Dr. Yukang Cao.* Jan 2024 – Aug 2024
 - Co-developed **GS-VTON** (IJCV 2026; equal contribution), an image-prompted 3D virtual try-on framework based on 3D Gaussian Splatting (3DGS).
 - Developed cross-view consistency mechanisms using personalized diffusion and attention-guided 3DGS editing to preserve appearance and geometry across viewpoints.
- Sharif University of Technology** Remote
 - *Under the supervision of Prof. Mohammad Hossein Rohban and Prof. Mahdieh Soleymani.* May 2022 – Jan 2024
 - Contributed to **RODEO** (ICML 2024), a data-centric framework for adversarially robust outlier detection using adaptive outlier exposure and adversarial training.
 - Developed and evaluated generative-model pipelines for synthesizing diverse, near-distribution OOD samples and improving robust outlier detection.
- Isfahan University of Technology** Isfahan, Iran
 - *B.Sc. thesis research under Prof. Abdolreza Mirzaei.* Jan 2023 – Dec 2023
 - Developed a community-detection method combining SDE-based diffusion models with graph neural networks.
 - Conducted model-design and experimental work aimed at improving identification of community structure in graph data.

PUBLICATIONS

- Exploiting Factored and Networked Structure for Scalable Equilibrium Computation in Markov Games** In preparation
Masoud Hadi, Chanwoo Park, Hossein Karshenas.
- GS-VTON: Controllable 3D Virtual Try-on with Gaussian Splatting** IJCV 2026
Yukang Cao*, Masoud Hadi*, Liang Pan, Ziwei Liu. *Equal contribution.
- DISTIL: Data-Free Inversion of Suspicious Trojan Inputs via Latent Diffusion** ICCV 2025
Hossein Mirzaei, Zeinab Sadat Taghavi, Sepehr Rezaee, Masoud Hadi, Moein Madadi, Mackenzie W. Mathis.
- RODEO: Robust Outlier Detection via Exposing Adaptive Out-of-Distribution Samples** ICML 2024
Hossein Mirzaei, Mohammad Jafari, Hamid Reza Dehbashi, Ali Ansari, Sepehr Ghobadi, Masoud Hadi, Arshia Soltani Moakhar, Mohammad Azizmalayeri, Mahdiah Soleymani Baghshah, Mohammad Hossein Rohban

HONORS AND AWARDS

- Ranked 37th out of approximately 15,000 participants in Iran's national M.Sc. Entrance Examination for Computer Engineering.
- Ranked among the top 0.5% in Iran's National University Entrance Examination.
- Received merit-based direct admission to Sharif University of Technology's M.Sc. program based on academic performance.
- Selected for transfer into the Computer Engineering major based on outstanding academic performance.

SELECTED COURSEWORK

[†]Graduate-level course

- **Mathematics & Theoretical Foundations:** Applied Linear Algebra (20/20), Probability and Statistics (19/20), Discrete Mathematical Structures (19/20), Game Theory (18/20), Stochastic Processes[†] (17/20)
- **Artificial Intelligence & Machine Learning:** Reinforcement Learning[†] (20/20), Machine Learning[†] (19.75/20), Basics of Machine Learning (18.2/20)

TEACHING EXPERIENCE

Discrete Mathematical Structures	<i>Fall 2022</i>
<i>Teaching Assistant under Prof. Abdolreza Mirzaei</i>	<i>Isfahan University of Technology</i>
Operating System Principles	<i>Spring 2023</i>
<i>Teaching Assistant under Prof. Zeynab Zali</i>	<i>Isfahan University of Technology</i>

TECHNICAL SKILLS

- **Programming:** Python, C/C++, MATLAB, R
- **ML & Scientific Computing:** PyTorch, Triton, TensorFlow, NumPy, pandas, scikit-learn, OpenCV
- **Tools:** Git, Linux

LANGUAGES

- Persian: Native
- English: TOEFL iBT 104 (R: 28, L: 26, S: 23, W: 27)